claiming that human consciousness is necessary to understand quantum mechanics, or that quantum mechanics may be necessary to understand consciousness. Much of this can be attributed to the impression that quantum mechanics is mysterious, and consciousness is mysterious, so maybe they have something to do with each other.

That's not wrong, as far as it goes. Maybe quantum mechanics and consciousness are somehow interconnected; it's a hypothesis we're welcome to contemplate. But according to everything we currently know, there is no good evidence this is actually the case.

Let's first examine whether quantum mechanics might help us understand consciousness. It's conceivable—though far from certain—that the rates of various neural processes in your brain depend on quantum entanglement in an interesting way, so that they cannot be understood by classical reasoning alone. But accounting for consciousness, as we traditionally think about it, isn't a straightforward matter of the rates of neural processes. Philosophers distinguish between the "easy problem" of consciousness—figuring out how we sense things, react to them, think about them—and the "hard problem"—our subjective, first-person experience of the world; what it is like to be us, rather than someone else.

Quantum mechanics doesn't seem to have anything to do with the hard problem. People have tried: Roger Penrose, for example, has teamed with anesthesiologist Stuart Hameroff to develop a theory in which objective collapse of the wave functions of microtubules in the brain helps explain why we experience consciousness. This proposal has not gained much acceptance in the neuroscience community. More important, it's unclear why it should matter for consciousness at all. It's perfectly conceivable that some subtle quantum processes in the brain, involving microtubules or something completely different, affect the rate at which our neurons fire. But this is of no help whatsoever in bridging the gap between "the firing of our neurons" and "our subjective, self-aware experience." Many scientists and philosophers, myself included, have no trouble believing that this gap is very bridgeable. But a tiny change in the rate of this or that neurochemical process doesn't seem to be relevant to understanding how. (And if it were, there's no reason the effect couldn't be repeated in nonhuman computers.)

Everettian quantum mechanics has nothing specific to say about the hard problem of consciousness that wouldn't be shared by any other view in